

DhatuChakra

(Dhatu = metal · Chakra = the wheel)

AI-Driven Life Cycle Assessment & Circularity

Engine for Indian Metals

Technical Documentation — Working Prototype v1

Smart India Hackathon (Internal Round) · Problem Statement #48 · Ministry of Mines

Repository: <https://github.com/grv-io/LCA-TOOL>

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1 Problem and Motivation

Life Cycle Assessment (LCA) is the standard method for quantifying the environmental footprint of metal production, and circularity indicators are becoming central to both national policy (National Critical Mineral Mission, circular-economy frameworks) and trade (the EU Carbon Border Adjustment Mechanism). In practice, however, LCA is out of reach for most Indian plants:

- a conventional study needs **25+ process parameters** most plants do not track;
- commercial inventory databases (e.g. ecoinvent) cost $\approx$ €4,000 per year;
- consultant-led studies take weeks and are not comparable across plants;
- global tools ignore Indian conditions (grid intensity, state-level variation, domestic data sources).

DhatuChakra answers this with a tool that needs only 5–8 inputs (or one plain sentence), fills the remaining parameters with a machine-learning imputer that reports its own confidence, and returns a full impact and circularity profile with quantified uncertainty — every number traceable to a cited public source.

2 System Overview

2.1 What the prototype does

1. **Scenario input** — metal (aluminium, steel, copper), production route, region or Indian state grid, recycled content, end-of-life recovery, renewable electricity share; optionally a plain-English description parsed by keyword matching.
2. **Parameter estimation** — a k -nearest-neighbours model predicts electricity and thermal-energy intensity with a real confidence score; every estimated field is visibly flagged and user-overridable.
3. **Impact computation** — GWP (kg CO₂e), cumulative energy (GJ), water (m³) and acidification (kg SO₂e) per tonne of metal, cradle-to-gate.
4. **Uncertainty** — a 1,000-run Monte Carlo simulation produces p05–p95 ranges and a histogram.
5. **Circularity** — the Material Circularity Indicator (Ellen MacArthur Foundation methodology), rendered as a 24-spoke chakra gauge.
6. **Decision support** — linear-vs-circular comparison with live sliders, benchmark strip (your plant vs India average vs world best practice), CBAM export-cost calculator, rule-based recommendations that cite the user’s own numbers, and an ISO-14044-style printable report.

2.2 Architecture

The prototype is a fully client-side web application (HTML/CSS/JavaScript, no backend, no external libraries; runs offline). The calculation engine, ML model and all data tables are transparent, inspectable files:

Module	Responsibility
js/data.js	All constants: routes, grids, benchmarks, CBAM, presets (single source of truth)
js/calcul.js	Engine: impact formulas, MCI, Monte Carlo, recommendations, text parser
js/ml_data.js	500 physics-generated training rows for the k -NN imputer
js/ui.js	Rendering: chakra gauge, Sankey, stage bar, histogram, benchmarks, Hindi toggle
5 HTML pages	Dashboard, Assess, Results, Compare, Report

Flow: *user input* $\rightarrow$ *parser* $\rightarrow$ *k-NN imputer (+confidence)* $\rightarrow$ *LCA engine* $\times$ *factor library* $\rightarrow$ *Monte Carlo* $\rightarrow$ *results, MCI, benchmarks, CBAM, recommendations, report*.

The full production build (designed, archived in the repository) replaces the factor engine with Brightway2.5, the k -NN with quantile-gradient-boosting imputation, and the keyword parser with an LLM parser — behind a FastAPI backend with PostgreSQL.

3 Calculation Methodology

3.1 Functional unit and boundary

The functional unit is **one tonne of metal at plant gate**; the system boundary is **cradle-to-gate**. The model is attributional.

3.2 Impact model

For each metal a linear (primary) and circular (secondary) route is parameterised by electricity intensity E (kWh/t), non-electric base emissions B (kg CO₂e/t) and base energy G (GJ/t). With recycled content $r \in [0, 1]$, renewable electricity share $s \in [0, 1]$ and grid intensity g_0 (kg CO₂e/kWh):

$$g_{\text{eff}} = g_0 (1 - s) + 0.04 s \quad (1)$$

$$\text{GWP} = (1 - r) (E_{\text{lin}} g_{\text{eff}} + B_{\text{lin}}) + r (E_{\text{circ}} g_{\text{eff}} + B_{\text{circ}}) \quad (2)$$

$$\text{Energy} = (1 - r) (0.0103 E_{\text{lin}} + G_{\text{lin}}) + r (0.0103 E_{\text{circ}} + G_{\text{circ}}) \quad (3)$$

$$\text{Water} = 0.004 \cdot \text{GWP} \quad \text{Acidification} = 0.005 \cdot \text{GWP} \quad (4)$$

where 0.04 is the life-cycle intensity of the renewable mix and 0.0103 GJ/kWh the primary-energy factor including generation losses. Route parameters:

Metal	Route	E (kWh/t)	B (kg CO ₂ e/t)	G (GJ/t)
Aluminium	Primary (Hall–Héroult)	14,500	6,300	50
Aluminium	Recycled (remelt)	800	350	8
Steel	BF–BOF	120	2,100	21
Steel	EAF (scrap)	450	380	2
Copper	Primary (pyromet.)	3,300	1,900	28
Copper	Recycled (secondary)	900	500	6

B for primary aluminium contains anode CO₂, PFC emissions, alumina refining thermal energy and mining/calcination/transport; for BF–BOF steel it contains coke, blast furnace and converter emissions (hence the small electricity term).

3.3 Grid intensity

National factors: India 0.71 (CEA CO₂ Baseline Database v19), world average 0.48, Europe 0.28 kg CO₂e/kWh. Additionally, ten *indicative* Indian state factors (derived from CEA v19 plus state generation mix) drive the interactive state map, from Himachal 0.18 (hydro) to Chhattisgarh 0.92 (coal). The same smelter moves from ≈ 8.9 to ≈ 19.1 t CO₂e/t between Himachal and Odisha — plant location is a climate decision.

3.4 Material Circularity Indicator

Following the Ellen MacArthur Foundation methodology (v3), with recycled feedstock fraction $F_r = r$, end-of-life collection rate C_r , recycling process efficiency $E_f = 0.9$ and utility factor $X = 1$ (mass normalised to $M = 1$):

$$V = 1 - F_r, \quad W_0 = 1 - C_r, \quad W_c = C_r(1 - E_f), \quad W_f = \frac{F_r(1 - E_f)}{E_f} \quad (5)$$

$$W = W_0 + \frac{1}{2}(W_f + W_c), \quad \text{LFI} = \frac{V + W}{2 + \frac{1}{2}(W_f - W_c)}, \quad \text{MCI} = \max(0, 1 - 0.9 \text{LFI}) \quad (6)$$

The UI renders MCI as an Ashoka-chakra gauge: $\text{round}(24 \cdot \text{MCI})$ of 24 spokes fill green.

3.5 Monte Carlo uncertainty

Each displayed range is the p05–p95 interval of 1,000 simulation runs. Per run, independent lognormal noise (geometric standard deviation 1.1) multiplies the grid factor, electricity intensity and base emissions: $x' = x \cdot \exp(\ln(1.1) Z)$, $Z \sim \mathcal{N}(0, 1)$ via Box–Muller. Water and acidification ranges derive from the GWP samples through their fixed ratios. The histogram is drawn from the same samples and is recomputed on every load.

3.6 k -NN parameter imputer

The “estimated” fields on the assessment page are predictions of a k -nearest-neighbours model, not hardcoded values.

- **Training data:** 500 rows generated from the engine’s own physics with $\pm 8\%$ noise (script `scripts/make_training_data.js`); regenerable at any time.
- **Features (normalised):** metal indicator, recycled content r , renewable share s , grid factor.
- **Prediction:** mean of the $k = 7$ nearest neighbours (Euclidean distance) for electricity and thermal intensity.
- **Confidence:** $1 - \sigma/\mu$ over the neighbours, clamped to $[0.50, 0.99]$ — it genuinely drops in uncertain regions (e.g. 94% at pure primary aluminium vs 78% at a 50% scrap blend).

The imputer is a suggestion layer: the engine itself always uses library physics values unless the user accepts or edits an estimate (auditable separation of AI and physics). Copper currently falls back to library defaults (model trained on Al/steel; retraining with copper is scheduled).

3.7 CBAM export-cost model

For EU-bound exports, annual CBAM exposure is estimated as

$$\text{Cost} = \max(0, \text{GWP}_t - b_m) \cdot \phi_y \cdot P_{\text{ETS}} \cdot X_{\text{EUR} \rightarrow \text{INR}} \cdot T \quad (7)$$

with indicative EU benchmark b_m (Al 6.0, steel 1.8 t CO₂e/t), phase-in factor ϕ_y per Regulation (EU) 2023/956 (2026: 2.5%, 2027: 5%, 2028: 10%, 2029: 22.5%, 2030: 48.5%, 2034: 100%), adjustable ETS price P_{ETS} (default €75/t) and export tonnage T . Example: a primary-aluminium smelter (16.6 t CO₂e/t) exporting 10,000 t/yr faces $\approx$ Rs. 1.8 crore in 2026, rising to $\approx$ Rs. 35 crore by 2030. Copper is outside the current CBAM product scope; the tool states this instead of showing a cost.

4 Validation

Engine outputs are pinned to an expected-output table; any deviation is treated as a bug. All six presets sit inside published ranges (IAI, worldsteel, EF 3.1 summaries):

Preset (India grid)	r	C_r	GWP (t CO ₂ e/t)	Energy (GJ/t)	Water (m ³ /t)	MCI
Aluminium — Primary	0.00	0.15	16.6	199	66	0.16
Aluminium — Recycled	1.00	0.70	0.92	16	3.7	0.83
Steel — BF–BOF	0.00	0.25	2.19	22	8.7	0.20
Steel — EAF	1.00	0.85	0.70	6.6	2.8	0.89
Copper — Primary	0.00	0.30	4.24	62	17.0	0.22
Copper — Recycled	1.00	0.80	1.14	15.3	4.6	0.87

Hand check (Al primary): $14,500 \times 0.71 + 6,300 = 16,595$ kg ≈ 16.6 t CO₂e/t — inside the IAI-consistent 16–20 range for coal-heavy grids. The recycled route is 94% lower, matching the widely cited $\approx 95\%$ saving. Reference ranges: recycled Al 0.5–1.5, BF–BOF 1.9–2.4, EAF 0.4–1.0 t CO₂e/t.

Benchmarks shown in the “Where you stand” card (indicative): Al primary — India average 17.8, world best practice (hydro smelters) 5.5; steel BF–BOF — India 2.55, best 1.8; copper primary — India 4.6, best 3.0 t CO₂e/t.

5 Data Sources

Factor	Source	Year
India grid intensity	CEA CO ₂ Baseline Database v19	2023
State grid intensities (indicative)	CEA v19 + state generation mix	2023
World/EU grid intensity	IEA emission factors	2023
Al smelting electricity, PFC, anode	International Aluminium Institute LCI	2022
Alumina refining and mining	EF 3.1 / ELCD	2022
Al remelting route	IDEMAT 2024, TU Delft	2024
BF–BOF and EAF inventories	worldsteel LCI methodology	2023
Copper routes (indicative)	EF 3.1 / IDEMAT 2024	2024
Transport factors	EF 3.1	2022
MCI methodology	Ellen MacArthur Foundation, MCI v3	2019
CBAM schedule and scope	Regulation (EU) 2023/956	2023

Every factor is public and free — the tool has **zero database licensing cost**, and each on-screen figure links back to its source (provenance table in the app and in the generated report).

6 Product Features (Prototype v1)

- **Dashboard** with six preset scenarios and a saved-scenario library (save/load/delete, persists locally).
- **Assess** — form + plain-English entry; interactive India state-grid map; k -NN estimated fields with live confidence badges; every estimate overridable.

- **Results** — four KPI cards with Monte-Carlo p05–p95 ranges and histogram; stage-contribution bar; material-flow Sankey with the recycling return loop; chakra MCI gauge; benchmark strip; CBAM calculator; provenance table.
- **Compare** — locked linear baseline vs live circular scenario; three circularity sliders; savings delta strip; recommendations that cite the current numbers.
- **Report** — ISO-14044-style document (goal & scope, inventory, results with uncertainty, circularity, recommendations, sources, limitations); print-to-PDF.
- **Hindi/English toggle** for the static UI (≈ 54 strings; full localisation planned).
- Fully offline-capable: fonts, map and all data ship with the page.

7 Limitations and Honest Caveats

1. The engine is a factor model with fixed water/acidification ratios — adequate for route comparison, not for site permitting. The production design swaps in a full matrix LCA (Brightway2.5).
2. State grid factors and copper/benchmark values are *indicative*, clearly labelled as such, pending plant-level data.
3. The k -NN imputer is trained on physics-generated (not measured) data; it demonstrates the imputation architecture and honest confidence reporting rather than field accuracy. Copper is not yet in the training set.
4. The plain-English parser is keyword-based in the prototype; the production build uses an LLM parser with validation.
5. Cradle-to-gate only; use-phase and end-of-life stages are parameterised in the MCI but not in the impact totals.

8 Roadmap (Full Build)

FastAPI + PostgreSQL backend; Brightway2.5 matrix engine; 100+ row cited factor library with regional fallback chains; LightGBM quantile imputer (p10/p50/p90) and surrogate model (target $R^2 > 0.95$, < 20 ms); LLM scenario parser and grounded recommendations; Monte-Carlo with per-factor pedigree uncertainty; multi-user scenario storage; full Hindi localisation; validation notebook against 8–10 published studies. The complete engineering plan, interface contracts and per-module specifications are archived in the repository ([docs/archive/](#)).

References

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DhatuChakra · every tonne, accounted for · built for Smart India Hackathon, Problem #48
Working prototype: all formulas, data and the ML model in this document are implemented and
verifiable in the repository.